Data Annotation

Data Science with Applications in Natural Language Processing

Learning goals

By the end, we will be able to:

- **explain** when and why human annotation is needed
- **choose** an annotation platform and format
- **design** a label scheme and guidelines for annotators
- **decide** who and how many annotators we need
- **organise** annotations in a table and find disagreements

Outline

1. What annotation is, and why
2. Annotation platforms
3. The manager: label scheme and guidelines
4. The annotators
5. In practice: annotations in a table

A question

“Great, another delay...”

Is it positive, negative or sarcastic?

- **Who** decides?
- Following which **guidelines**?
- What do we do when they **disagree**?

Data annotation



- The model learns to become a **machine annotator**
- It labels like humans do, with little or no intervention

Also called labelling, rating or coding.

When do we need humans?

Objective task	Subjective task
which language is the text in?	is the comment toxic?
does it contain a URL?	is it sarcastic?
is there a traffic light in the image?	what emotion does it express?

- In subjective tasks there is no **single** right answer for everyone

Annotation is not mechanical

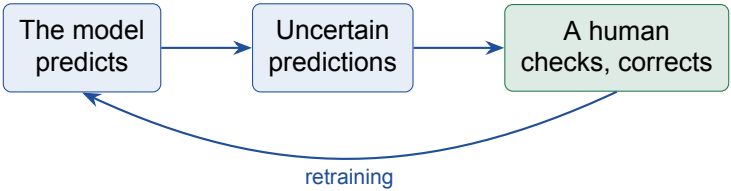
This is where human knowledge enters the system

Labels capture:

- human **judgements**
- **values**
- and **disagreements**

Whatever goes into the labels, the model will learn.

Human-in-the-loop



- Judgement is not given once, at the start
- It becomes a **continuous feedback loop**

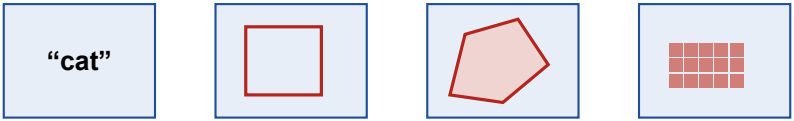
What a platform does

- An environment where annotators assign labels, following **guidelines**
- Dozens of options, commercial and open source

The choice depends on

- the kind of data: text, image, audio
- how complex the annotation is
- the number of annotators
- cost and time

Annotating images



e.g. VGG Image Annotator (University of Oxford).

Annotating text

- A **class** for a whole sentence or document

  I do not trust the system → **negative**

- **Spans** inside the text: entities, relations, parts of speech

  Achilles^{PER} left Troy^{LOC}

  e.g. WebAnno.

Is a spreadsheet enough?

Yes, for

- small datasets
- pilot experiments
- teaching

No, for

- large scale
- text spans, boxes on images
- many annotators

Row = item · columns = text, labels, metadata. A table, just like a DataFrame.

The manager of an experiment



E.g. the goal: automatic sentiment analysis of texts, or film ratings.

The form of a label

Categories

positive · negative · neutral

e.g. sentiment analysis

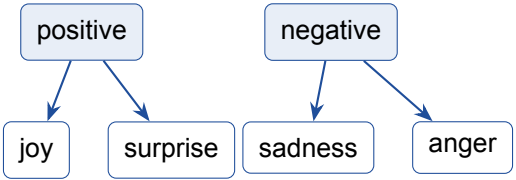
Scale

★★★★☆ (1 to 5)

e.g. film reviews

- A scale has an **order**: 4 is closer to 5 than 1 is
- Categories do not

Level of detail



- More detail: richer information
- But **harder** annotation, usually less agreement

One label or several?

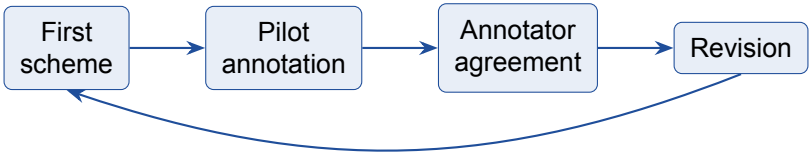
“I’m glad she left, but I will miss her”

Single-label
joy **or** sadness: we must choose

Multi-label
joy **and** sadness: a mixed state

```
labels = pd.Series(["joy|sadness", "anger", "joy"])
labels.str.get_dummies("|")     # one column per label, 0 or 1
```

A scheme is designed, not found



- Decisions follow the **phenomenon** and the **goal**
- The pilot shows whether the scheme captures the complexity

Guidelines for annotators

- **Definitions** for every label
- **Examples** and **counter-examples**
- **Borderline cases** and how to handle them
- What do I do when I'm not sure?

Written for someone reading about the problem **for the first time**.

Example guideline: toxicity

Definition

Toxic: a rude or insulting comment that may make someone leave the discussion.

Comment	**Label**
“You’re an idiot if you believe that”	toxic
“I strongly disagree with the measure”	not toxic
“Another ‘brilliant’ idea from the ministry”	borderline: sarcasm, not a personal insult

A hypothetical example; borderline cases are the most useful part of any guidelines.

Publish the guidelines

Together with the data and the results

- **Transparency**: how the labels came about
- **Reproducibility**: others can repeat the experiment
- **Interpretation**: what exactly the model was learning

Quality and disagreement

- **Who** and **how many** annotators per item
- **Quality control**:
  - items with a known answer (gold)
  - repeating the same item to the same annotator
- A **rule** for disagreement: majority, discussion, or keep them all

Annotation is not an add-on: it is a **central design stage** of a machine learning system.

Who annotates

	Quality	**Cost**	**Scale**
Experts	high	high	small
Non-experts (crowd)	needs checking	low	large, fast
Mixed teams	experts validate	medium	medium

Experts: doctors, lawyers, linguists. Non-experts rely mainly on the guidelines.

Training the annotators

Guidelines are not enough on their own

- Understanding the **goal**
- Examples and counter-examples
- A pilot round to resolve questions

Less random variation, better coordination between annotators.

One annotator or many?

One is enough

- very simple tasks
- “is the text in Greek?”

Many are needed

- subjective tasks
- to measure **agreement**
- to estimate noise and reliability

What influences an annotator

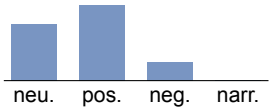
- cultural and social background
- personal experience
- **fatigue**

In sentiment, offensive or political language, differences are not errors: they are **human diversity**.

One label or a distribution?

- If annotators agree: one label
- If opinions are **polarised**: keep the distribution

Recall the Iliad corpus: each verse had annotator shares per sentiment, not a single label.



The first verse of the Iliad.

Two annotators in one table

```
df = pd.DataFrame({
    "text":   [f"text {i}" for i in range(10)],
    "ann_a":  ["pos","neg","pos","neu","neg","pos","neu","neg","pos","neu"],
    "ann_b":  ["pos","neg","neu","neu","neg","pos","neg","neg","pos","neu"],
    "conf_a": [3, 2, 1, 3, 3, 2, 1, 3, 3, 2],
    "conf_b": [3, 2, 1, 2, 3, 3, 1, 2, 3, 2]})
```

- One label column and one **confidence** column (1–3) per annotator
- Wide format: one annotator per column

Do they agree?

```
df["agree"] = df["ann_a"] == df["ann_b"]
df["agree"].mean()                # 0.8
df[~df["agree"]]
```

text	ann_a	ann_b	conf_a	conf_b
text 2	pos	neu	1	1
text 6	neu	neg	1	1

- Agreement on 8 out of 10
- In both disagreements, both annotators were **unsure**

Where do they disagree?

```
pd.crosstab(df["ann_a"], df["ann_b"])
```

ann_a \ ann_b	neg	neu	pos
neg	**3**	0	0
neu	1	**2**	0
pos	0	1	**3**

- The diagonal: agreement
- Disagreements always sit **next to** neutral, never positive vs negative

Disagreement and confidence

```
df["min_conf"] = df[["conf_a", "conf_b"]].min(axis=1)
df.groupby("min_conf")["agree"].mean()
# 1: 0.0    2: 1.0    3: 1.0
```

- When either annotator is unsure: **no** agreement
- Low confidence may mean:
  - an ambiguous item
  - insufficient guidelines
  - a hard category

It does not necessarily mean the annotator acted in bad faith.

Many annotators: long format

```
long = df.melt(id_vars="text", value_vars=["ann_a", "ann_b"],
               var_name="annotator", value_name="label")
long.shape     # (20, 3): one row per (text, annotator)
```

- Platforms usually export in **long** format
- pivot back to wide, for comparison
- groupby("text") gives the label distribution per text

Raw agreement is not enough

If 90% of the texts are “neutral”...

two annotators answering **at random** with the same proportions agree on average

$$0.9^2 + 0.1^2 = 0.82$$

We need measures that correct for chance. More under: agreement estimation.

Summary

1. Annotation brings human judgement into the system
2. Platform and format follow the data and the task
3. Schemes and guidelines are designed, piloted and published
4. Many annotators for subjective tasks; disagreement is information
5. In a table: agreement, crosstab, confidence, long format

Exercise 1: design an experiment

For toxicity detection in comments, define:

1. the label **scheme** (categories or a scale? one label or several?)
2. the **guidelines**, with at least one borderline case
3. how many **annotators** per item, and who
4. how you check **quality**
5. what you do with **disagreements**

Exercise 1: one possible answer

Scheme	toxic / ok, plus optionally *who* is targeted (multiple choice)
Guidelines	a definition, 5–10 examples; borderline: irony, quoting an insult to condemn it
Annotators	3 per comment, of different ages and regions; confidence 1–3
Quality	a 50-comment pilot, 10% gold items, 5% repeated for self-consistency
Disagreements	keep the **distribution**; revise the guidelines if agreement is low

There is no single right answer: what matters is the reason behind each choice.

Exercise 2: two annotators, eight comments

```
df = pd.DataFrame({
  "text":   [f"comment {i}" for i in range(1, 9)],
  "ann_a":  ["toxic","ok","ok","toxic","ok","toxic","ok","ok"],
  "ann_b":  ["toxic","ok","toxic","toxic","ok","ok","ok","toxic"],
  "conf_a": [3, 3, 1, 3, 2, 2, 3, 1],
  "conf_b": [3, 3, 2, 2, 3, 1, 3, 1]})
```

1. Add an agree column and compute the percent agreement
2. Which comments are disagreements?
3. Are disagreements related to low confidence?

Exercise 2: solution

```
df["agree"] = df["ann_a"] == df["ann_b"]
df["agree"].mean()                    # 0.625
df.loc[~df["agree"], "text"]          # comment 3, 6, 8
df["min_conf"] = df[["conf_a", "conf_b"]].min(axis=1)
df.groupby("min_conf")["agree"].mean()    # 1: 0.0  2: 1.0  3: 1.0
```

- All three disagreements have **one** annotator at confidence 1
- In two of the three, B says toxic: pd.crosstab shows it
- With 8 comments this is a **hint**, not a conclusion

For discussion

1. All your annotators are students from the same city. Which texts are at risk of being labelled “wrong”?
2. When would you prefer two experts to twenty non-experts?
3. One annotator is often unsure. Do you remove them, or change the guidelines?